# Thesis Title

Thesis Subtitle




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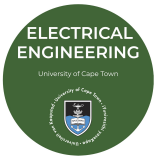
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**August 2026**

BSc (Eng) final year project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and Computer Engineering in the Department of Electrical Engineering at the University of Cape Town

*Keywords:* keyword1; keyword2; keyword3



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**Word count** [run latexmk] words (excluding preamble, appendices and bibliography)
**Page count** 11 pages (excluding preamble, appendices and bibliography)



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University of Cape Town

Fri 7 August, 2026

# Abstract

**Thesis Title**

Joab Gray **Kloppers**

*Fri 7 August, 2026*

# Acknowledgements

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# List of Symbols

| Symbol | Description | Unit |
|---|---|---|
| $\boldsymbol{B}$ | Magnetic flux density | T |
| $\boldsymbol{E}$ | Electric field | V/m |
| $\boldsymbol{J}$ | Current density | $A/m^2$ |
| $t$ | Time | s |
| $\varepsilon_0$ | Permittivity of free space ($8.85 \times 10^{-12}$) | $C^2/N{\cdot}m^2$ |
| $\mu_0$ | Permeability of free space ($4\pi \times 10^{-7}$) | $N/A^2$ |

# Chapter 1

# Introduction

testing something [1] testing something else

this is more content

## 1.1 Background

### 1.1.1 Subsection

In ullamcorper blandit quam. Sed tortor felis, mattis at, pretium et, tempus sit amet, pede. Mauris laoreet scelerisque dui. Curabitur eu risus. Aliquam gravida vehicula erat. Pellentesque suscipit ultricies dui. Aliquam faucibus sagittis felis. Nulla enim. Nulla facilisi. Sed non purus luctus magna bibendum pretium. Aliquam egestas mattis sapien. Duis laoreet nisl at est. Integer at massa. Duis ut sapien sit amet ipsum pretium eleifend. Vivamus augue ante, pretium non, ornare ac, egestas non, lectus. Phasellus tristique. Nullam nec quam id neque pellentesque ornare. Maecenas semper elit id dui. Fusce libero urna, porttitor a, molestie in, vestibulum sit amet, enim. Morbi facilisis nulla at purus elementum pharetra. Proin malesuada, odio eu tempus posuere, diam justo consequat.*

#### Sub-subsection

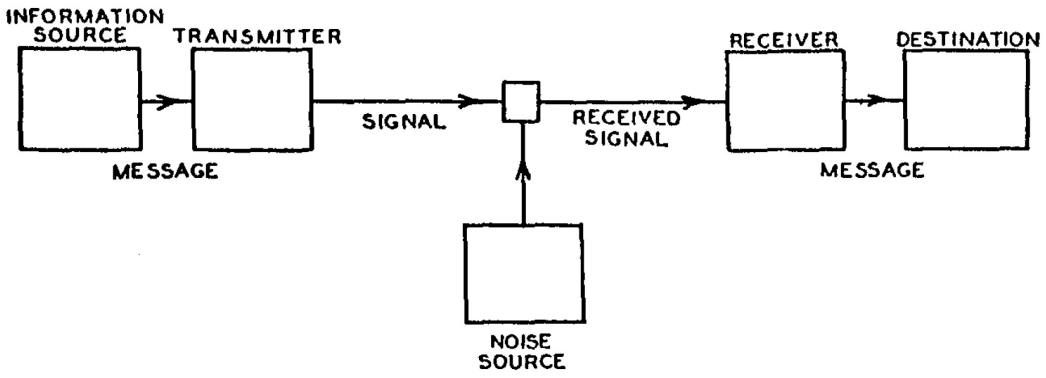



**Figure 1.1: A simplified block diagram of a general communication system.** A communication system can be abstracted as follows: an *information source* produces a message to be sent through a *transmitter* which converts it to a signal which can be sent via a communication *channel*. The received signal is noisy and is converted back to the message by the *receiver* which is then delivered to the *destination*. Taken from [2].

*This is a footnote

**Table 1.1: Table title.** Extended description of tables contents.

| **Heading 1** | **Heading 2** | |
|---|---|---|
| | *Subheading 1* [units] | *Subheading 2* [units] |

## 1.2 Problem statement

Equations are numbered and include the chapter number. As they are considered part of the text, they include punctuation, and each parameter must be described in the text below. Ensure that the units are stated and where possible use SI units. For example, the Maxwell equations are expressed as:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}, \tag{1.1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{1.2}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{1.3}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}, \tag{1.4}$$

where $\boldsymbol{B}$ is the magnetic flux density in T, $\boldsymbol{J}$ is the total current density in A/m$^2$, $\boldsymbol{E}$ is the electric field in V/m, $\mu_0$ is the permeability of free space ($4\pi \times 10^{-7}$ H/m), and $\epsilon_0$ is the permittivity of free space (8.85 $\times 10^{-12}$ C$^2$/N·m$^2$).

## 1.3 Project objectives

The Global Positioning System (GPS) is used for positioning. The Marginal Ice Zone (MIZ) is of particular interest in the Southern Ocean (SO).

## 1.4 Scope, limitations and assumptions

## 1.5 Plan of development

# Chapter 2

# Literature review

Textual, where the author is the subject of the sentence: **(author?)** [2] established the information-theoretic limits that still bound modern communication systems.

Parenthetical, at the end of a clause: The effect is well documented [3].

Several sources at once, which natbib sorts and compresses: The same conclusion is reached by three independent routes [3–5].

A page, chapter or section within a source: The register map is given in full [6, Ch. 8].

The year alone, where the sentence already names the author: Shannon's channel capacity theorem [**(year?)**] is the origin of the result.

# Chapter 3

# Theoretical development

# Chapter 4

# Design description

# Chapter 5

# Experimental design

# Chapter 6

# Results

# Chapter 7

# Discussion

# Chapter 8

# Conclusions

# Chapter 9

# Further work

# Appendix A

# GitHub repository

This appendix provides a link to the GitHub repository where the code and additional materials for this final year project report can be found.

## A.1 Title for appendix A.1

- GitHub Repository: Repo name

# Appendix B: Ethics approval

# Appendix C: Topic description

The original topic description used during the project selection process is included below. This topic has since been further refined, and the exact problem that was investigated is described in Section 1.2.

# Appendix D: ECSA Graduate Attributes

## Project mapping to ECSA Graduate Attributes

The table below provides a structured reflection on where each of the required Graduate Attributes (GAs) as defined by Engineering Council of South Africa (ECSA) associated with the final year project is met. This includes reference to the specific Chapter/Section (where applicable).

| GA | Graduate Attribute Description | Student reflection: How this outcome was achieved during the project |
|---|---|---|
| GA1 | *Problem solving* | |
| GA4 | *Investigations, experiments and data analysis* | |
| GA5 | *Use of engineering tools* | |
| GA6 | *Professional and technical communication* | |
| GA8 | *Individual and collaborative teamwork* | |
| GA9 | *Independent learning ability* | |

# Bibliography


[1] "Camera Depth Testing Methodology."

[2] C. E. Shannon, "A mathematical theory of communication," *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[3] A. N. Surname and B. C. Othersurname, "Title of the journal article," *Name of the Journal*, vol. 12, no. 4, pp. 101–117, 2024.

[4] A. N. Surname, *Title of the Book*, 3rd ed. City, Country: Name of the Publisher, 2020.

[5] A. N. Surname and B. C. Othersurname, "Title of the conference paper," in *Proceedings of the Name of the Conference*, City, Country, 2023, pp. 1–6.

[6] Name of the Organisation, "Title of the page or document," https://example.org/path/to/document, 2025, accessed: 2025-02-11.